

HFP: Empirical Companion Note (v2.2)

Verified engineering claims and multi-seed recall experiments for an $O(1)$ -memory causal language-model architecture

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Abstract

This note is the empirical companion to the Hyper-Flux Projection (HFP) theory preprint hosted in this OSF project. The theory is *inspiration* for the architecture, not evidence; conversely, nothing here validates the physics. All claims below are machine-learning claims, tested at small scale ($\leq 1\text{M}$ parameters, synthetic recall tasks, CPU), multi-seed where stated, and reproducible from the public repository: <https://github.com/kayra-hn/HFP> (see `RESULTS.md` and `review_scripts/`). Chance level is 3.3% throughout. Effect *patterns* are the results; absolute numbers are scale-dependent. No language-model benchmark claims are made.

1 Architecture in one paragraph

HFP pairs windowed local attention with a per-layer recurrent linear-attention memory ($M \in \mathbb{R}^{D \times H}$, $z \in \mathbb{R}^D$) whose inference-time state is constant in context length. Selectable axes: retention law (**exp** geometric decay vs. **cubic_flux**, an exact discretization of $d\theta/d\tau = -\eta\theta^3$ giving state-magnitude-dependent decay $\lambda_t = 1/\sqrt{1 + 2\eta z_t^2}$); write rule (**additive** vs. DeltaNet-style **delta**); and key feature map (**elu** vs. DPFP, key dimension $4\times$). Engineering verified by independent review: chunk-consistency of all mode combinations, causal correctness, $O(1)$ state, and exactness of the cubic discretization and of its two-pass parallel form.

2 Methodological finding: supervision density gates learnability

Single-query recall sequences (one supervised token per sequence) sit at the $\ln(\text{vocab})$ loss plateau and never learn once the context exceeds ~ 300 tokens, at any tested learning rate or curriculum—although the same model learns the same task easily with 8 queries per sequence. **Retention claims measured with sparse-supervision tasks are optimization artifacts.** All experiments below use dense multi-query sequences.

3 Retention law and write rule (3 seeds, context 160)

Mean accuracy (%) over seeds $\{0, 1, 2\}$; 600 steps; gap buckets in tokens.

configuration	1–15	16–47	48–95	96+
exp + additive (baseline)	44.4	32.7	18.8	8.1
exp + delta write	52.1	31.1	16.4	6.6
cubic_flux + additive	31.3	23.0	11.8	4.7

`cubic_flux` trails the exponential baseline at this scale and is seed-fragile (one of three seeds failed to learn); it is parked as a long-horizon hypothesis. Delta writes help only at short range (see §6).

4 Length generalization (3 seeds) — main positive result

Models trained at context 160 are evaluated, unchanged, at $2\text{--}8\times$ the training length. Fixed-gap accuracy *increases* with evaluation length in all three seeds (gap <48 bucket shown; %):

seed	eval 160	320	640	1280
0	38.2	63.2	71.7	75.0
1	32.4	39.3	45.5	42.9
2	40.0	54.7	75.4	85.7

The apparent “training-length cliff” (failure to *train* at context ≥ 320 on a fixed budget) is an optimization artifact, not an architectural limit: the $O(1)$ recurrent state directly supports **train-short** $\rightarrow$ **infer-long** deployment.

5 The memory is interference-limited, not decay-limited

Holding length fixed (context 640) and scaling the number of stored facts $P = 8 \rightarrow 16 \rightarrow 24$ monotonically degrades fixed-gap accuracy in all three seeds. This explains §4: longer streams at fixed fact count have lower fact density, hence less interference.

6 Capacity axis (DPFP) — first clear mechanism win

DPFP (Schlag et al., 2021; key dimension $4\times$) attacks exactly the interference limit. Confirmed across 3 seeds: at context 640, gap 256+, $P=8$, the baseline is at chance in all seeds {5.4, 3.2, 2.5} while DPFP reaches {10.7, 12.9, 31.1}; in the highest-interference cell ($P=24$, gap 128–255) baseline 5.3 vs. DPFP 13.2. DPFP also removes the baseline’s weak-seed instability (seed-1 peak accuracy 39% $\rightarrow$ 95%). It compounds with length generalization (*single seed*, seed 2, trained at 160, evaluated at 1280; multi-seed confirmation of this compound cell is a standing open item, §8; %):

variant @1280	<48	48–127	128–255	256+
elu (baseline)	85.7	69.4	29.9	5.4
dpfp	88.1	87.8	70.1	33.5

Delta writes do *not* help on this task family: the interference is cross-key feature overlap (a capacity problem), not same-key overwriting.

7 Current recipe and honest ledger

Recommended configuration: **exp** decay, **additive** writes, **dpfp** features, standard FFN, dense multi-query training at short context, deployment at arbitrary length. Parked or negative: **cubic_flux** (behind baseline at tested scale; exact parallel form implemented for a future targeted long-horizon test); two-tier consolidation memory (prototype verified, fair evaluation pending); all single-seed results are labeled as such in `RESULTS.md`.

8 Standing open items (not yet claimed)

The following are scripted and reproducible but not yet run at the scale needed to make a claim; none of the results above depends on them.

- **External family baseline.** A pure-PyTorch Gated Linear Attention (GLA) baseline on the identical dense-retention task (`review_scripts/baseline_compare.py`) provides an out-of-family comparison point. Until it is run at matched parameters, all comparisons in this note are HFP-internal (retention law / write rule / feature map), not versus the efficient-recurrent literature.
- **Pure-memory ablation.** Windowed attention can also see the last `max_short_len` ring-buffer tokens in the streaming path, so “information travels only through M, z ” is verified cleanly only with the ring buffer disabled (`review_scripts/pure_memory_ablation.py`, `ring buffer = 1`).
- **Multi-seed DPFP $\times$ length.** The compound DPFP + train-short / infer-long cell at 1280 (§7) is currently single-seed; seeds 0–1 pending.
- **Language-model benchmark.** No LM-benchmark claim is made; all results are architecture-level synthetic recall.

Reproducibility and license

Code: <https://github.com/kayra-hn/HFP> (AGPL-3.0). Model card: <https://huggingface.co/kayrahan35/HFP-01-Memory-Model>. Reproduction commands are listed in `RESULTS.md`; regression tests in `smoke_test.py` and `review_scripts/verify_claims.py`.